Trustworthy uncertainty quantification for quantitative and diffusion body MRI, and its decision-use in MR-guided adaptive radiotherapy: a field review

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Abstract

Quantitative and diffusion body MRI increasingly delivers not an image but a *number* — a tissue parameter offered to a clinical decision — and, in MR-guided adaptive radiotherapy, a number that can steer dose. This review asks whether the *uncertainty* attached to that number is trustworthy enough to act on. We organise the field along the spine a clinician implicitly walks: *trust* (is the error bar honest?), *value of information* (does a trusted bar change the decision?), and *action* (what happens when it is deployed to drive dose?), over a foundations layer (what is measured, and how) and beneath a gap map. Synthesising the literature across estimation, calibration, decision analysis, and MR-guided radiotherapy, we find that trust is conditional: broadly defensible for the diffusion coefficient D and ADC, but fragile for the perfusion parameters f and D^*, where measured test–retest reliability is poor and where deep-learning uncertainty quantification typically reports only marginal, aggregate calibration. The decision-analytic apparatus to price a calibrated interval — net benefit, decision calibration, value of information — is mature but rarely connected to quantitative-MRI uncertainty; and the propagation of *biomarker* uncertainty

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into the adaptive dose decision remains the immature link. We close with four open problems sited at the seams between trust, value, and action. This review reports no new measurement; its contribution is synthesis, taxonomy, and gap identification.
Keywords: quantitative MRI, intravoxel incoherent motion, uncertainty quantification, calibration, conformal prediction, value of information, MR-guided adaptive radiotherapy, dose painting

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1. Scope and framing

Quantitative and diffusion body MRI — intravoxel incoherent motion (IVIM), diffusion-weighted ADC/DKI, dynamic contrast-enhanced (DCE) perfusion, and relaxometry — increasingly produces not just an image but a *number*: a tissue parameter offered to a clinical decision. The question this review surveys is whether the **uncertainty** attached to that number is trustworthy enough to act on, and what happens when it is used to steer **MR-guided adaptive radiotherapy** (MR-Linac / MRgART).

We organise the literature along the spine a clinician implicitly walks:

1. **Trust** — is the error bar honest (calibrated, covering)?
2. **Value of information** — given a trusted bar, does it change the decision?
3. **Action** — when deployed to drive dose, where does it hold and where does it break?

with a **foundations** layer beneath (what is measured, and how) and a **gap map** above. The survey covers UQ for quantitative/diffusion body MRI and its decision-use in adaptive RT; it excludes non-quantitative MRI, neuro-only work, generic computer-vision UQ, and non-MR-guided RT (the scope boundary is recorded in the repository's `ASSUMPTIONS.md`). Table 1 lays out the survey axis and the thematic buckets it induces.

Table 1: The survey axis. The field is classified by the question each work answers about *somebody's* uncertainty estimate; a cross-cutting gap map (Sec. 6) sits over the three axes at the seams between them.

axis	what the bucket asks
Foundations	What is the estimand, and what estimator (with what uncertainty) produced it? (Sec. 2)
Trust	Is the error bar honest — calibrated and covering? (Sec. 3)
Value of information	Given a trusted bar, does it change an oncologic decision? (Sec. 4)
Action	When deployed to drive dose on an MR-linac, where does it hold and break? (Sec. 5)
Gap map	Where, at the seams between the axes, do the field's UQ-trust questions remain open? (Sec. 6)

2. Foundations: estimands and estimators

Before honesty can be assessed, the estimand and estimator must be fixed. The IVIM model itself was introduced to separate molecular diffusion from capillary-network microcirculation within a voxel [1, 2], treating randomly oriented capillary flow as a pseudo-diffusion process [3]; DCE-MRI quantification rests on the compartmental transfer-constant model [4] and its standardised quantities K^{trans}, v_e, k_{ep} [5].

The estimator is not innocent. Cramér–Rao/Fisher analysis shows the achievable precision of IVIM D and f is strongly SNR-dependent [6], and the same bounds can be inverted to design SNR-efficient acquisitions [7]. Empirically, IVIM parameter estimates differ *significantly* across fitting algorithms in the upper abdomen, with Bayesian fitting giving the lowest inter-reader and inter-subject variability [8]; simulation [9] and pancreatic-cancer data [10] corroborate the Bayesian advantage for D and f. A parallel line replaces hand-built fits with learned ones — neural-network IVIM/kurtosis mapping

[11], supervised and unsupervised physics-informed deep fitting [12, 13], and self-supervised forward-model estimation [14] — with recent work decomposing predictive uncertainty into aleatoric and epistemic components [15]. DCE shares the trajectory, e.g. Bayesian spatial tracer-kinetic estimation [16]. **Take-away:** the uncertainty a method reports is inseparable from the estimator that produced it, so trust must be assessed per-estimator, not per-biomarker.

3. Trust: is the error bar honest?

3.1. The general machinery of trust

The UQ community offers a toolkit for honest uncertainty that the imaging literature draws on. Distribution-free **conformal prediction** wraps any black-box predictor in a set/interval valid at a user-chosen error rate [17, 18]. **Calibration** methods recalibrate regression intervals post-hoc [19] and correct the systematic over-confidence of modern deep networks [20]. Practical deep UQ families — deep ensembles [21], MC-dropout as approximate Bayesian inference [22], and single-pass evidential regression [23] — give usable uncertainty, but their quality **degrades under dataset shift**, so trust must be tested against shift, not only in-distribution [24].

3.2. The metrology of trust in quantitative imaging

Imaging has its own standards for honest reliability. The QIBA framework anchors technical performance on three metrology areas — linearity/bias, repeatability, reproducibility [25] — with companion guidance on comparing algorithms [26] and on consistent terminology [27]. Reliability reporting leans on the intraclass correlation coefficient [28] and on limits-of-agreement rather than correlation for method comparison [29].

3.3. Where trust is empirically tested in body MRI

When these standards are applied to body diffusion/perfusion MRI, the error bars are often **not** honest — and the dishonesty is parameter- and cohortdependent. ADC is a comparatively reliable biomarker but its in-vivo reproducibility is markedly worse than phantom reproducibility [30], and the body-DWI consensus already urged baseline reproducibility studies as part of study design [31, 32]. The perfusion parameters are the weak point: short-term test–retest repeatability coefficients for IVIM f and D^* reach $\sim$126% and $\sim$197% in rectal cancer [33], D^* reproducibility is only moderate (ICC ≈ 0.55, CV $\approx 20\%$) [34], and multi-scanner liver IVIM within-subject CVs run from $\sim$5% for D up to $\sim$14% for perfusion-fraction components [35]. Cross-modal anchoring does not rescue D^*: its correlation with DCE K^{trans} is weak in one rectal cohort ($r = 0.389$) [36] and non-significant in another [34]. **Takeaway:** trust in body-MRI UQ is conditional — broadly defensible for D/ADC, fragile for f/D^* — yet most deep-learning UQ reports aggregate or marginal calibration [15], leaving the conditional failure region under-characterised.

4. Value of information: does a trusted bar change the decision?

A calibrated interval earns its cost only if it moves an action. The decision-analytic literature supplies the apparatus: **decision curve analysis / net benefit** evaluates a model by clinical net benefit across the decision threshold [37–39], where the threshold encodes the clinician's relative weighting of a missed disease against an unnecessary intervention; performance frameworks argue such decision-analytic measures should be reported whenever a model supports decisions [40], and relative-utility curves quantify how much achievable utility a predictor captures [41]. From the inference side, **decision/loss-calibrated** methods tie the posterior approximation to the downstream decision — decision calibration [42] and loss-calibrated Bayesian inference [43, 44]. From health economics, **value-of-information** analysis (EVPI) prices a measurement by combining the probability it changes a decision with the benefit of that change [45], under the decision-theoretic stance that choices follow expected net benefit rather than statistical significance [46]. **Takeaway:** the machinery to ask "does this body-MRI interval change an oncologic decision?" exists and is mature —

but is rarely connected to quantitative-MRI uncertainty.

5. Action: deployment in MR-guided adaptive radiotherapy

The action setting where a quantitative-MRI biomarker most directly drives dose is the MR-Linac. Integrating diagnostic-quality MRI with a linear accelerator [47, 48] matured into a clinically feasible 1.5 T MRI-guided system [49, 50] and an emerging paradigm for adaptive radiation oncology [51, 52]. The biological premise is **dose painting** — prescribing a non-uniform dose from a biological target volume derived from functional/molecular images [53–55] — with DWI/ADC a leading candidate response biomarker [56] and quantitative MRI proposed to update the plan to observed tumour response [51, 57]. Two cautions recur. First, **uncertainty must be carried into the optimisation**: robust/probabilistic planning incorporates motion and uncertainty directly rather than relying on PTV margins [58]. Second, **technical validation gates clinical use**: imaging biomarkers on an MR-linac require repeatability/reproducibility validation before driving decisions, because the integrated scanner differs from a diagnostic system [59]. **Takeaway:** the field is converging on biomarker-driven adaptive dose, but the propagation of *biomarker uncertainty* into the dose decision is the immature link.

6. Gap map: open problems

The field's unresolved UQ-trust questions cluster at the **seams between the axes** — which is also, stated as honest positioning rather than as this review's organising centre, where the author's own program sits.

- **G1 — conditional honesty (trust seam).** Repeatability evidence shows f/D^* are far less reliable than D/ADC [33–35], yet deep-learning IVIM-UQ typically reports marginal/aggregate calibration [15]. *Open:* calibration conditioned on the regime where it fails, and conformal/coverage guarantees that respect it [18, 24].

- **G2 — from calibration to decision (trust→value seam).** The decision-analytic toolkit [37, 42] is rarely applied to quantitative-MRI uncertainty. *Open:* when does a calibrated body-MRI interval actually change an oncologic decision versus merely being reported?

- **G3 — uncertainty into dose (value→action seam).** Adaptive-RT reviews call for technically validated biomarkers to steer dose [51, 59], and robust planning can ingest uncertainty [58]. *Open:* end-to-end propagation of *biomarker* uncertainty (not just geometric/motion uncertainty) into the adaptive dose decision, and the identifiability walls that cap it.

- **G4 — estimator-dependence of trust (cross-cutting).** Reported uncertainty depends on the estimator [8, 12]. *Open:* standards that make UQ comparable across classical and learned estimators, extending QIBA metrology [25, 27] to the uncertainty itself, not only the point estimate.

7. Honest scope and discussion

This review's contribution is the **synthesis, taxonomy, and gap map** above — not new data. It asserts no experimental result; every empirical statement is attributed to a verified external source. Where the author's own program is relevant it is named only as positioning within G1–G4, never as the organising thread.

As a portfolio artifact the review is deliberately *trigger-independent*: it strengthens a first research application by demonstrating field command and supplying a citable synthesis, and it depends on none of the author's own results publishing. In this role it **absorbs** what an earlier plan called a separate portfolio-thickening "Buttress" note: rather than a stand-alone companion, the field command and the open-problem map are folded here, and the gap map (Sec. 6) doubles as the positioning of the author's program within the field's open seams — honest positioning, not the survey's organising centre. The four gaps are stated as the *field's* open problems; the author's contributions, where they exist, sit inside G1–G4 on equal footing with external work and are not asserted here as published results.

The dominant failure mode for a review of this kind is the phantom or mis-quoted citation. The repository makes that mechanical: every entry in the citation base carries a resolvable identifier (DOI, arXiv eprint, or stable proceedings URL) and a one-line verified claim, and a citation gate fails the build on any unresolved identifier, any orphan between the bibliography and the verified-claim ledger, or any `\cite` in this manuscript with no backing entry. Four reliability/oncology foundations are flagged at thesis level: their central claim is title/abstract-evident, but any load-bearing *number* from them is re-pulled verbatim from the source before use rather than approximated.

CRediT authorship contribution statement

Avery Karlin (sole author): Conceptualization; Methodology; Software (citation gate and build); Formal analysis; Investigation; Data curation; Writing — original draft; Writing — review and editing. Terms follow the NISO CRediT taxonomy.

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Data availability

This review reports no new experimental data. The verified citation base (`limbo.bib`), the per-citation verified-claim ledger (`CITATIONS.md`), and the citation gate (`verify_citations.py`) that enforces resolvable identifiers and zero phantom citations are openly available in the project repository; one command (`./reproduce.sh`) re-runs the gate, the test-suite, and the manuscript build.

Declaration of Generative AI and AI-assisted technologies in the writing process

Generative AI use disclosure: during the preparation of this work the author used Claude (Anthropic), including Claude Code, for code scaffolding (the citation gate and build), manuscript drafting assistance, and copy-editing. After using these tools the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

[1] D. Le Bihan, E. Breton, D. Lallemand, P. Grenier, E. Cabanis, M. Laval-Jeantet, Mr imaging of intravoxel incoherent motions: application to diffusion and perfusion in neurologic disorders, Radiology 161 (1986) 401–407. doi:10.1148/radiology.161.2.3763909.

[2] D. Le Bihan, E. Breton, D. Lallemand, M. L. Aubin, J. Vignaud, M. Laval-Jeantet, Separation of diffusion and perfusion in intravoxel incoherent motion mr imaging, Radiology 168 (1988) 497–505. doi:10.1148/radiology.168.2.3393671.

[3] D. Le Bihan, What can we see with ivim mri?, NeuroImage 187 (2019) 56–67. doi:10.1016/j.neuroimage.2017.12.062.

[4] P. S. Tofts, A. G. Kermode, Measurement of the blood-brain barrier permeability and leakage space using dynamic mr imaging. 1. fundamental concepts, Magnetic Resonance in Medicine 17 (1991) 357–367. doi:10.1002/mrm.1910170208.

[5] P. S. Tofts, G. Brix, D. L. Buckley, J. L. Evelhoch, E. Henderson, M. V. Knopp, H. B. Larsson, T. Y. Lee, N. A. Mayr, G. J. Parker, R. E.

Port, J. Taylor, R. M. Weisskoff, Estimating kinetic parameters from dynamic contrast-enhanced t1-weighted mri of a diffusable tracer: standardized quantities and symbols, Journal of Magnetic Resonance Imaging 10 (1999) 223–232. doi:10.1002/(SICI)1522-2586(199909)10:3<223::AID-JMRI2>3.0.CO;2-S.

[6] Q. Zhang, Y.-X. J. Wang, H. T. Ma, J. Yuan, Cramér-rao bound for intravoxel incoherent motion diffusion weighted imaging fitting, in: 35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2013, pp. 511–514. doi:10.1109/EMBC.2013.6609549.

[7] O. Jalnefjord, M. Montelius, G. Starck, M. Ljungberg, Optimization of b-value schemes for estimation of the diffusion coefficient and the perfusion fraction with segmented intravoxel incoherent motion model fitting, Magnetic Resonance in Medicine 82 (2019) 1541–1552. doi:10.1002/mrm.27826.

[8] S. Barbieri, O. F. Donati, J. M. Froehlich, H. C. Thoeny, Impact of the calculation algorithm on biexponential fitting of diffusion-weighted mri in upper abdominal organs, Magnetic Resonance in Medicine 75 (2016) 2175–2184. doi:10.1002/mrm.25765.

[9] P. T. While, A comparative simulation study of bayesian fitting approaches to intravoxel incoherent motion modeling in diffusion-weighted mri, Magnetic Resonance in Medicine 78 (2017) 2373–2387. doi:10.1002/mrm.26598.

[10] O. J. Gurney-Champion, R. Klaassen, M. Froeling, S. Barbieri, J. Stoker, M. R. W. Engelbrecht, J. W. Wilmink, M. G. Besselink, A. Bel, H. W. M. van Laarhoven, A. J. Nederveen, Comparison of six fit algorithms for the intra-voxel incoherent motion model of diffusion-weighted magnetic resonance imaging data of pancreatic cancer patients, PLOS ONE 13 (2018) e0194590. doi:10.1371/journal.pone.0194590.

[11] M. Bertleff, S. Domsch, S. Weingärtner, J. Zapp, K. O'Brien, M. Barth, L. R. Schad, Diffusion parameter mapping with the combined intravoxel

incoherent motion and kurtosis model using artificial neural networks at 3 t, NMR in Biomedicine 30 (2017) e3833. doi:10.1002/nbm.3833.

[12] S. Barbieri, O. J. Gurney-Champion, R. Klaassen, H. C. Thoeny, Deep learning how to fit an intravoxel incoherent motion model to diffusion-weighted mri, Magnetic Resonance in Medicine 83 (2020) 312–321. doi:10.1002/mrm.27910.

[13] M. P. T. Kaandorp, S. Barbieri, R. Klaassen, H. W. M. van Laarhoven, H. Crezee, P. T. While, A. J. Nederveen, O. J. Gurney-Champion, Improved unsupervised physics-informed deep learning for intravoxel incoherent motion modeling and evaluation in pancreatic cancer patients, Magnetic Resonance in Medicine 86 (2021) 2250–2265. doi:10.1002/mrm.28852.

[14] S. D. Vasylechko, S. K. Warfield, O. Afacan, S. Kurugol, Self-supervised ivim dwi parameter estimation with a physics based forward model, Magnetic Resonance in Medicine 87 (2021) 904–914. doi:10.1002/mrm.28989.

[15] N. Casali, A. Brusaferri, G. Baselli, S. Fumagalli, E. Micotti, G. Forloni, R. Hussein, G. Rizzo, A. Mastropietro, A comprehensive framework for uncertainty quantification of voxel-wise supervised models in ivim mri, arXiv preprintAlso NMR in Biomedicine, 2026 (2025). arXiv:2508.04588.

[16] V. J. Schmid, B. Whitcher, A. R. Padhani, N. J. Taylor, G.-Z. Yang, Bayesian methods for pharmacokinetic models in dynamic contrast-enhanced magnetic resonance imaging, IEEE Transactions on Medical Imaging 25 (2006) 1627–1636. doi:10.1109/TMI.2006.884210.

[17] G. Shafer, V. Vovk, A tutorial on conformal prediction, Journal of Machine Learning Research 9 (2008) 371–421.
URL `https://www.jmlr.org/papers/v9/shafer08a.html`

[18] A. N. Angelopoulos, S. Bates, A gentle introduction to conformal prediction and distribution-free uncertainty quantification, arXiv preprint (2021). arXiv:2107.07511.

[19] V. Kuleshov, N. Fenner, S. Ermon, Accurate uncertainties for deep learning using calibrated regression, in: Proceedings of the 35th International Conference on Machine Learning (ICML), Vol. 80, 2018, pp. 2796–2804. URL https://proceedings.mlr.press/v80/kuleshov18a.html

[20] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning (ICML), Vol. 70, 2017, pp. 1321–1330. URL https://proceedings.mlr.press/v70/guo17a.html

[21] B. Lakshminarayanan, A. Pritzel, C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems 30 (NeurIPS), 2017. arXiv:1612.01474.

[22] Y. Gal, Z. Ghahramani, Dropout as a bayesian approximation: representing model uncertainty in deep learning, in: Proceedings of the 33rd International Conference on Machine Learning (ICML), Vol. 48, 2016, pp. 1050–1059. arXiv:1506.02142.

[23] A. Amini, W. Schwarting, A. Soleimany, D. Rus, Deep evidential regression, in: Advances in Neural Information Processing Systems 33 (NeurIPS), 2020. arXiv:1910.02600.

[24] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, J. Snoek, Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems 32 (NeurIPS), 2019. arXiv:1906.02530.

[25] D. L. Raunig, L. M. McShane, G. Pennello, C. Gatsonis, P. L. Carson, J. T. Voyvodic, R. L. Wahl, B. F. Kurland, A. J. Schwarz, M. Gönen, G. Zahlmann, M. V. Kondratovich, K. O'Donnell, N. Petrick, P. E. Cole, B. Garra, D. C. Sullivan, Quantitative imaging biomarkers: a review of statistical methods for technical performance assessment, Statistical Methods in Medical Research 24 (2015) 27–67. doi:10.1177/0962280214537344.

[26] N. A. Obuchowski, H. X. Barnhart, A. J. Buckler, G. Pennello, X.-F. Wang, J. Kalpathy-Cramer, H. J. Kim, A. P. Reeves, Statistical issues in the comparison of quantitative imaging biomarker algorithms using pulmonary nodule volume as an example, Statistical Methods in Medical Research 24 (2015) 107–140. doi:10.1177/0962280214537392.

[27] D. C. Sullivan, N. A. Obuchowski, L. G. Kessler, D. L. Raunig, C. Gatsonis, E. P. Huang, M. Kondratovich, L. M. McShane, A. P. Reeves, D. P. Barboriak, A. R. Guimaraes, R. L. Wahl, Metrology standards for quantitative imaging biomarkers, Radiology 277 (2015) 813–825. doi:10.1148/radiol.2015142202.

[28] T. K. Koo, M. Y. Li, A guideline of selecting and reporting intraclass correlation coefficients for reliability research, Journal of Chiropractic Medicine 15 (2016) 155–163. doi:10.1016/j.jcm.2016.02.012.

[29] J. M. Bland, D. G. Altman, Statistical methods for assessing agreement between two methods of clinical measurement, The Lancet 327 (1986) 307–310. doi:10.1016/S0140-6736(86)90837-8.

[30] M. M. Jafar, A. Parsai, M. E. Miquel, Diffusion-weighted magnetic resonance imaging in cancer: reported apparent diffusion coefficients, in-vitro and in-vivo reproducibility, World Journal of Radiology 8 (2016) 21–49. doi:10.4329/wjr.v8.i1.21.

[31] D.-M. Koh, D. J. Collins, Diffusion-weighted mri in the body: applications and challenges in oncology, American Journal of Roentgenology 188 (2007) 1622–1635. doi:10.2214/AJR.06.1403.

[32] A. R. Padhani, G. Liu, D.-M. Koh, T. L. Chenevert, H. C. Thoeny, T. Takahara, A. Dzik-Jurasz, B. D. Ross, M. Van Cauteren, D. Collins, D. A. Hammoud, G. J. S. Rustin, B. Taouli, P. L. Choyke, Diffusion-weighted magnetic resonance imaging as a cancer biomarker: consensus and recommendations, Neoplasia 11 (2009) 102–125. doi:10.1593/neo.81328.

[33] H. Sun, Y. Xu, Q. Xu, K. Shi, W. Wang, Rectal cancer: short-term reproducibility of intravoxel incoherent motion parameters in 3.0t magnetic resonance imaging, Medicine (Baltimore) 96 (2017) e6866. doi:10.1097/MD.0000000000006866.

[34] X. Yang, X. Xiao, B. Lu, Y. Chen, Z. Wen, S. Yu, Perfusion-sensitive parameters of intravoxel incoherent motion mri in rectal cancer: evaluation of reproducibility and correlation with dynamic contrast-enhanced mri, Acta Radiologica 60 (2019) 569–577. doi:10.1177/0284185118791201.

[35] G. Simchick, T. J. Allen, D. Hernando, Reproducibility of intravoxel incoherent motion quantification in the liver across field strengths and gradient hardware, Magnetic Resonance in Medicine 92 (2024) 2652–2669. doi:10.1002/mrm.30237.

[36] H. Sun, Y. Xu, Q. Xu, J. Duan, H. Zhang, T. Liu, L. Li, Q. Chan, S. Xie, W. Wang, Correlation between intravoxel incoherent motion and dynamic contrast-enhanced magnetic resonance imaging parameters in rectal cancer, Academic Radiology 26 (2019) e134–e140. doi:10.1016/j.acra.2018.08.012.

[37] A. J. Vickers, E. B. Elkin, Decision curve analysis: a novel method for evaluating prediction models, Medical Decision Making 26 (2006) 565–574. doi:10.1177/0272989X06295361.

[38] A. J. Vickers, B. Van Calster, E. W. Steyerberg, Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests, BMJ 352 (2016) i6. doi:10.1136/bmj.i6.

[39] A. J. Vickers, B. van Calster, E. W. Steyerberg, A simple, step-by-step guide to interpreting decision curve analysis, Diagnostic and Prognostic Research 3 (2019) 18. doi:10.1186/s41512-019-0064-7.

[40] E. W. Steyerberg, A. J. Vickers, N. R. Cook, T. Gerds, M. Gonen, N. Obuchowski, M. J. Pencina, M. W. Kattan, Assessing the performance of predic-

tion models: a framework for traditional and novel measures, Epidemiology 21 (2010) 128–138. doi:10.1097/EDE.0b013e3181c30fb2.

[41] S. G. Baker, N. R. Cook, A. Vickers, B. S. Kramer, Using relative utility curves to evaluate risk prediction, Journal of the Royal Statistical Society, Series A (Statistics in Society) 172 (2009) 729–748. doi:10.1111/j.1467-985X.2009.00592.x.

[42] S. Zhao, M. P. Kim, R. Sahoo, T. Ma, S. Ermon, Calibrating predictions to decisions: a novel approach to multi-class calibration, in: Advances in Neural Information Processing Systems 34 (NeurIPS), 2021. arXiv:2107.05719.

[43] S. Lacoste-Julien, F. Huszár, Z. Ghahramani, Approximate inference for the loss-calibrated bayesian, in: Proceedings of the 14th International Conference on Artificial Intelligence and Statistics (AISTATS), Vol. 15, 2011, pp. 416–424.
URL `https://proceedings.mlr.press/v15/lacoste_julien11a.html`

[44] M. P. Vadera, S. Ghosh, K. Ng, B. M. Marlin, Post-hoc loss-calibration for bayesian neural networks, in: Proceedings of the 37th Conference on Uncertainty in Artificial Intelligence (UAI), Vol. 161, 2021. arXiv:2106.06997.

[45] J. C. Felli, G. B. Hazen, Sensitivity analysis and the expected value of perfect information, Medical Decision Making 18 (1998) 95–109. doi:10.1177/0272989X9801800117.

[46] K. Claxton, The irrelevance of inference: a decision-making approach to the stochastic evaluation of health care technologies, Journal of Health Economics 18 (1999) 341–364. doi:10.1016/S0167-6296(98)00039-3.

[47] J. J. W. Lagendijk, B. W. Raaymakers, A. J. E. Raaijmakers, J. Overweg, K. J. Brown, E. M. Kerkhof, R. W. van der Put, B. Hardemark, M. van Vulpen, U. A. van der Heide, Mri/linac integration, Radiotherapy and Oncology 86 (2008) 25–29. doi:10.1016/j.radonc.2007.10.034.

[48] B. W. Raaymakers, J. J. W. Lagendijk, J. Overweg, J. G. M. Kok, A. J. E. Raaijmakers, E. M. Kerkhof, R. W. van der Put, I. Meijsing, S. P. M. Crijns, F. Benedosso, M. van Vulpen, C. H. W. de Graaff, J. Allen, K. J. Brown, Integrating a 1.5 t mri scanner with a 6 mv accelerator: proof of concept, Physics in Medicine and Biology 54 (2009) N229–N237. doi:10.1088/0031-9155/54/12/N01.

[49] B. W. Raaymakers, I. M. Jürgenliemk-Schulz, G. H. Bol, M. Glitzner, A. N. T. J. Kotte, B. van Asselen, J. C. J. de Boer, J. J. Bluemink, S. L. Hackett, M. A. Moerland, S. J. Woodings, J. W. H. Wolthaus, H. M. van Zijp, M. E. P. Philippens, R. Tijssen, J. G. M. Kok, E. N. de Groot-van Breugel, I. Kiekebosch, L. T. C. Meijers, C. N. Nomden, G. G. Sikkes, P. A. H. Doornaert, W. S. C. Eppinga, N. Kasperts, L. G. W. Kerkmeijer, J. H. A. Tersteeg, K. J. Brown, B. Pais, P. Woodhead, J. J. W. Lagendijk, First patients treated with a 1.5 t mri-linac: clinical proof of concept of a high-precision, high-field mri guided radiotherapy treatment, Physics in Medicine and Biology 62 (2017) L41–L50. doi:10.1088/1361-6560/aa9517.

[50] J. J. W. Lagendijk, B. W. Raaymakers, M. van Vulpen, The magnetic resonance imaging–linac system, Seminars in Radiation Oncology 24 (2014) 207–209. doi:10.1016/j.semradonc.2014.02.009.

[51] R. Otazo, P. Lambin, J.-P. Pignol, M. E. Ladd, H.-P. Schlemmer, M. Baumann, H. Hricak, Mri-guided radiation therapy: an emerging paradigm in adaptive radiation oncology, Radiology 298 (2021) 248–260. doi:10.1148/radiol.2020202747.

[52] W. A. Hall, E. Paulson, X. A. Li, B. Erickson, C. Schultz, A. Tree, M. Awan, D. A. Low, B. A. McDonald, T. Salzillo, C. K. Glide-Hurst, A. U. Kishan, C. D. Fuller, Magnetic resonance linear accelerator technology and adaptive radiation therapy: an overview for clinicians, CA: A Cancer Journal for Clinicians 72 (2022) 34–56. doi:10.3322/caac.21707.

[53] C. C. Ling, J. Humm, S. Larson, H. Amols, Z. Fuks, S. Leibel, J. A.

Koutcher, Towards multidimensional radiotherapy (md-crt): biological imaging and biological conformality, International Journal of Radiation Oncology, Biology, Physics 47 (2000) 551–560. doi:10.1016/S0360-3016(00)00467-3.

[54] S. M. Bentzen, Theragnostic imaging for radiation oncology: dose-painting by numbers, The Lancet Oncology 6 (2005) 112–117. doi:10.1016/S1470-2045(05)01737-7.

[55] S. M. Bentzen, V. Gregoire, Molecular imaging-based dose painting: a novel paradigm for radiation therapy prescription, Seminars in Radiation Oncology 21 (2011) 101–110. doi:10.1016/j.semradonc.2010.10.001.

[56] S. Leibfarth, R. M. Winter, H. Lyng, D. Zips, D. Thorwarth, Potentials and challenges of diffusion-weighted magnetic resonance imaging in radiotherapy, Clinical and Translational Radiation Oncology 13 (2018) 29–37. doi:10.1016/j.ctro.2018.09.002.

[57] P. J. van Houdt, Y. Yang, U. A. van der Heide, Quantitative magnetic resonance imaging for biological image-guided adaptive radiotherapy, Frontiers in Oncology 10 (2021) 615643. doi:10.3389/fonc.2020.615643.

[58] J. Unkelbach, M. Alber, M. Bangert, R. Bokrantz, T. C. Y. Chan, J. O. Deasy, A. Fredriksson, B. L. Gorissen, M. van Herk, W. Liu, H. Mahmoudzadeh, O. Nohadani, J. V. Siebers, M. Witte, H. Xu, Robust radiotherapy planning, Physics in Medicine and Biology 63 (2018) 22TR02. doi:10.1088/1361-6560/aae659.

[59] P. J. van Houdt, H. Saeed, D. Thorwarth, C. D. Fuller, W. A. Hall, B. A. McDonald, MR-Linac Consortium Imaging Biomarker Working Group, Integration of quantitative imaging biomarkers in clinical trials for mr-guided radiotherapy: conceptual guidance for multicentre studies from the mr-linac consortium imaging biomarker working group, European Journal of Cancer 153 (2021) 64–71. doi:10.1016/j.ejca.2021.04.041.